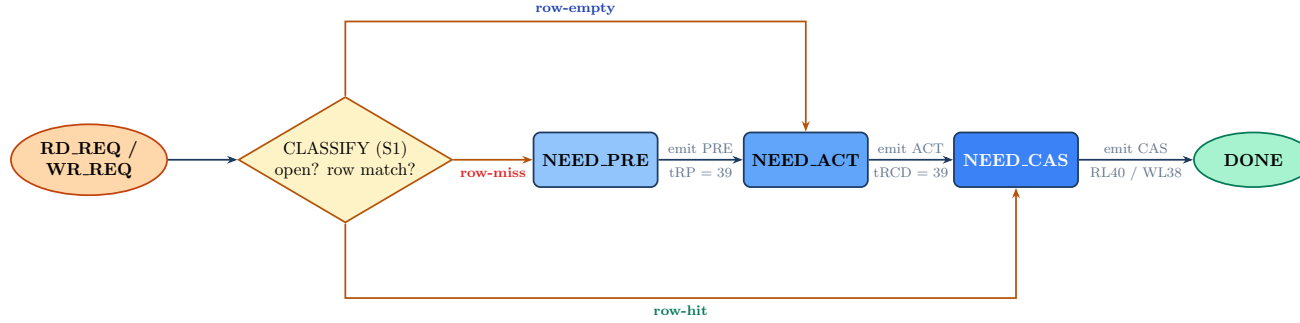


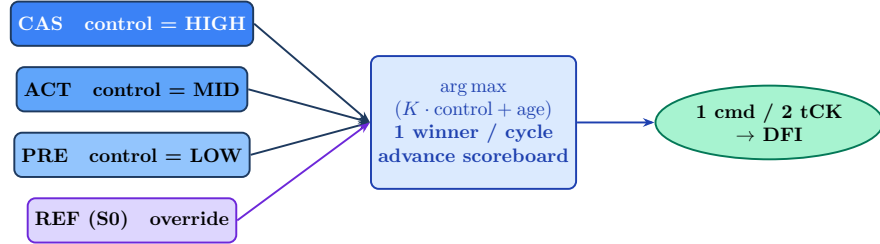
Per-Bank Request FSM – RMC DDR5 Scheduler

One bank shown; N_BANKS copies run independently, merged by the weight arbiter (1 cmd / 2tCK). Timing = b4800 bin (tCK 0.4167ns). Companion to scheduler_bank_fsm.md / scheduler_staged_logic.md.



Read and write share the FSM; only the CAS flavour (CAS_RD/CAS_WR) and the tail timing differ (write adds tWR recovery before PRE). **No marching token** — every NEED_* slot nominates in parallel; the arbiter emits one.

Weight arbiter — emission control



Aging: age[lane] += 1 every waiting cycle (incl. timing-blocked); = 0 on win. Starved lane climbs until it preempts — bounded starvation.

△ **Scaling caveat (weights pass).** Age ticks *every* waiting cycle, so without the K scale it swamps the fixed control weight and the arbiter degenerates to oldest-first (loses CAS-first). K , control values, and AGE_MAX = one joint knob-set.

Classify — the fork

bank state	case	chain	work_state
closed / idle	row-empty	ACT→CAS	NEED_ACT
open, row==R	row-hit	CAS	NEED_CAS
open, row!=R	row-miss	PRE→ACT→CAS	NEED_PRE

Latency chains (tCK, b4800)

case	chain	first data
row-hit (<i>best</i>), rd	CAS	RL = 40
row-hit, wr	CAS	WL = 38
row-empty, rd	ACT→CAS	39+40 = 79
row-miss (<i>worst</i>), rd	PRE→ACT→CAS	39+39+40 = 118
+ full burst	+ BL/2	+8 = 126
row-miss + refresh	...REF...→PRE→ACT→CAS	+tRFC = +708
write tail → PRE	CAS_WR → ... → PRE	WL+BL2+tWR = 118
R→W turnaround	CAS_RD→CAS_WR	tRTW = 12
W→R turnaround	CAS_WR→CAS_RD	+tWTR_L = 70

$L_{\text{miss}} = 126\text{tCK}$, service = BL/2 = 8tCK/burst $\Rightarrow$ latency floor $N = L_{\text{miss}}/8 \approx 16$ in-flight reads to hide one row-miss. Buffer depth = 64 (margin). See staged_logic §0.

N_BANKS FSMs run in parallel; cross-bank tCCD_L / tRRD_L / tFAW / DQ apply at the arbiter, not inside a bank. Paths $\neq$ emissions: CA bus is 1 cmd / 2tCK.